

Failure Modes and Fault Signatures

Robot Weld 01 (robot-weld-01) | Sensor 1: AXIS_4_SERVO_TORQUE

Machine	Robot Weld 01 (machine ID robot-weld-01, six-axis spot-welding robot, Body shop, site vit-lab)
Sensor	Sensor 1: AXIS_4_SERVO_TORQUE, "Axis 4 servo torque", sensor ID ROBOT-WELD-01.AXIS_4_SERVO_TORQUE, type torque, unit Nm
Component / subsystem	Robot forearm (axis 4): axis 4 servo drive and gearbox; planned agent component tag robot_forearm_axis_4; suggested twin mesh 105-six-axis-welding-robot/arm-swing
Document title	Failure Modes and Fault Signatures
Document ID	RW01-S1-DOC-02
Revision / date	1.0 / 2026-10-08
Purpose	Explains how physical, electrical, thermal, lubrication, control and sensor-related faults appear in Sensor 1 (axis 4 servo torque) telemetry, so that a diagnostic agent can move from an abnormal pattern to candidate faults and then separate competing candidates with numerical discriminators.
Scope	Faults that can change AXIS_4_SERVO_TORQUE of Robot Weld 01: the project fault GEARBOX_WEAR [P] and eleven engineering-reference faults [E][I]. Normal levels are in Document 01; procedures and decision matrices in Document 03; maintenance in Document 04; worked cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_SENSOR1.md (catalog limits, simulation equations, topics, tests).
[D]	Derived value: computed by the document authors from the repository's own simulation equations (noise-free unless stated).
[E]	Engineering reference: general industrial practice for robot servo-axis diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. An ESP32 gateway (or a software simulator) plays the robot controller and generates values from documented equations. The limits in this document that are tagged [P] are illustrative constants chosen for the demo in the repository, not OEM specifications. Values tagged [E] and [I] are not manufacturer-certified.

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1. Purpose and Method

Document 02 is the fault-signature library for Sensor 1. Its job is to answer: given this torque pattern, which faults could have produced it, what separates them, and what must be inspected. It uses the same terminology as Document 01 (cycle mean M, severity index S, harmonic amplitude A6, noise sigma, per-cycle peak) and the same provenance tags.

1.1 What is project fact and what is engineering reference

Only one fault is defined in the repository for Robot Weld 01: GEARBOX_WEAR, a damaged gear mesh of the axis 4 gearbox, with exact equations for torque and for the tool center point deviation [P]. All other faults in this document are engineering-reference faults of a robot servo axis [E]. Their magnitudes are illustrative [I], expressed in the same Sensor 1 features so that they can be compared numerically with the project fault, and they are intended to be replaced by real data. The simulator cannot produce them; they matter because a diagnostic agent must be able to say that an observed pattern is not GEARBOX_WEAR.

1.2 Feature vector used to describe every fault

Table 1. Signature features

Feature	Symbol	Definition	Healthy value	Use
Mean shift	dM	Window mean minus baseline mean (13.0 Nm)	0 +/- 0.5 Nm	Size of baseline change
Swing change	dSw	Change of the 1x cycle amplitude (7.0 Nm healthy)	0	Separates load change from friction/wear
Sixth-harmonic ripple	A6	Amplitude at 0.75 Hz (6x cycle)	0 to 0.2 Nm	Gear-mesh signature in this dataset
Noise ratio	kN	Residual sigma divided by 0.45 Nm	1.0 (0.9 to 1.2)	Roughness
Phase-locked bump	B(phase)	Residual to template concentrated in a phase window	none	Local mechanical event
Onset type	-	step (within one cycle), ramp (minutes to hours), drift (days or longer), intermittent	-	Fault dynamics
TCP response	dT	Change of TOOL_CENTER_POINT_DEVIATION mean (healthy 0.104 mm)	0	Corroboration, kinematic accuracy
Gun temperature response	dG	Change of WELD_GUN_TEMP (healthy 37 to 58 degC)	0	Excludes thermal/process cause
Idle offset	dI	Torque while axis is holding position, versus baseline hold	0	Detects offset or brake faults

2. Failure-Mode Classification

Table 2. Failure-mode classification for Sensor 1

ID	Fault	Class	Affected component	Basis	Sensor 1 sensitivity	Frequency of occurrence [E]
F01	Gearbox wear: damaged gear mesh (GEARBOX_WEAR)	Mechanical wear	Axis 4 gearbox	[P]	High: primary indicator	Common in high-duty axes
F02	Lubrication degradation or loss	Lubrication	Axis 4 gearbox grease / oil	[E]	High for mean, low for ripple	Common
F03	Brake drag (partially engaged holding brake)	Mechanical / electrical	Axis 4 motor brake	[E]	High for mean and idle torque	Occasional
F04	Bearing degradation (motor, output, forearm bearings)	Mechanical wear	Axis 4 bearings	[E]	Medium: noise and slow mean	Occasional
F05	Backlash growth or mechanical looseness (gear backlash, coupling, mounting bolts)	Mechanical / alignment	Gearbox-to-arm interface, mounting	[E]	Medium: reversal spikes, little mean	Occasional
F06	External drag: cable dress pack, hose, spatter on seals	Mechanical / installation	Dress pack, seals	[E]	Medium: phase-locked bump	Common
F07	Overload, payload increase or collision	Overload	Forearm, tool, gun	[E]	High: step or spike	Occasional
F08	Drive or current-measurement fault (offset, gain, phase imbalance)	Electrical	Axis 4 servo drive, current sensors	[E]	High: offset, aliasing	Rare
F09	Thermal effect (motor or gearbox overtemperature, cold viscosity)	Thermal	Motor winding, gearbox oil	[E]	Medium: reversible drift	Common
F10	Control-system, program or tuning change	Control	Controller, program, speed override	[E]	High: shape change	Very common (not a fault)
F11	Sensor signal fault: flatline, saturation, offset, spike train	Sensor / signal	Torque estimate path	[E]	High	Rare
F12	Data-path fault: stale, sequence gap, clock error, identity conflict	Telemetry	ESP32 gateway, broker, backend	[P] (data-path table)	Corrupts interpretation	Common in prototypes

2.1 Reading the signature space

Gearbox wear in this dataset is the only fault that raises the cycle mean, adds a sixth-harmonic ripple and raises the noise at the same time, in fixed proportions. Friction faults raise the mean without the ripple; electrical and sensor faults produce offsets or flat traces that persist also when the arm is stationary; mechanical events produce phase-locked bumps or spikes; load changes alter the swing. Figure 1 shows the model spectrum; Figure 2 shows representative shapes; Figure 3 (Section 5) shows how the corroborating TCP deviation develops.

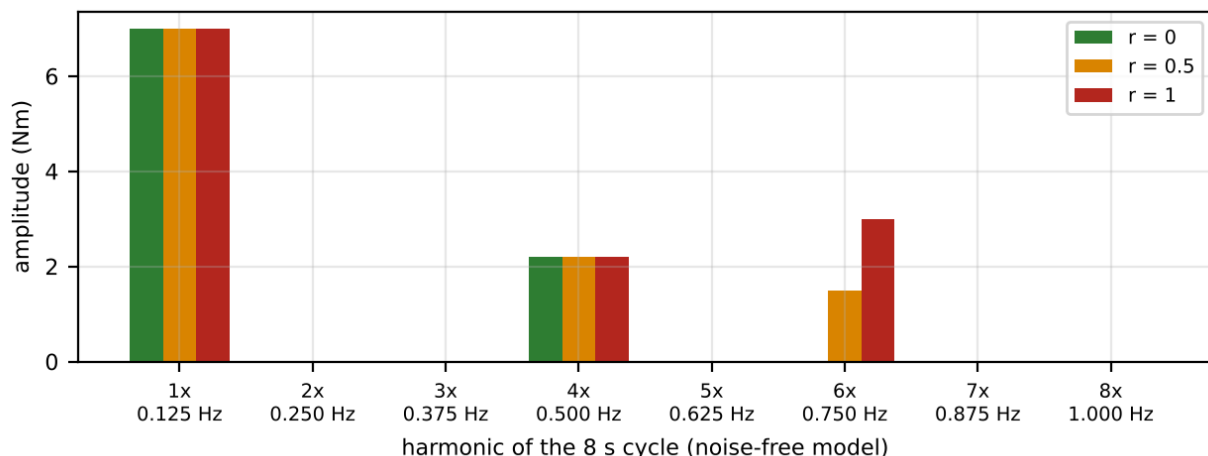


Figure 1. Harmonic content of Sensor 1 by gearbox-wear severity: the 6x line (0.75 Hz) grows as $3.0 r$ Nm while the 1x and 4x lines stay constant [D].

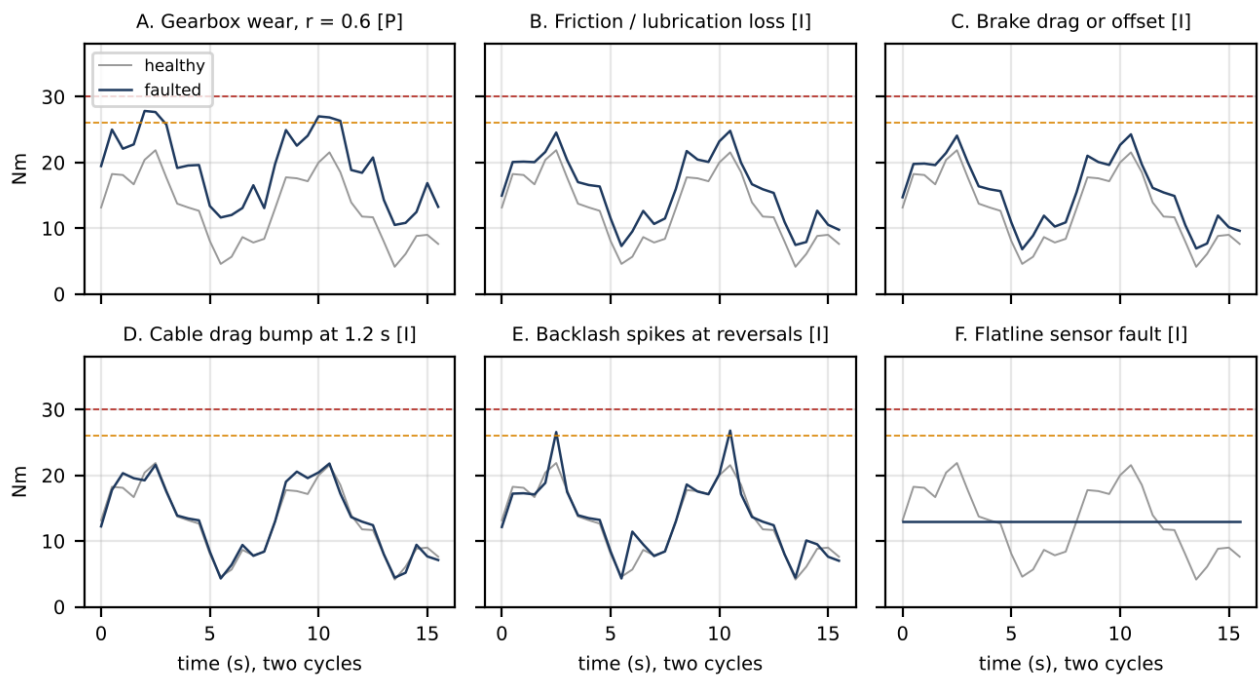


Figure 2. Representative cycle shapes for Sensor 1: A is the project fault ($r = 0.6$); B to F are illustrative engineering-reference shapes [I] over two cycles with healthy trace in grey.

3. Fault Signatures

Table 3. Fault signature summary (cycle features relative to healthy baseline)

Fault	dM (Nm)	dSw	A6 (Nm)	Noise kN	Onset	Phase pattern	dT (mm)	dG	dI (idle)
F01 Gearbox wear [P]	+10.5 r (0 to +10.5)	0	+3.0 r (0 to 3.0)	1 + 1.5 r (1 to 2.5)	ramp (rampSec 5 to 600 s)	Whole cycle, peak lobes 1.5 to 3.5 s	+0.55 r (0 to +0.55)	0	Not modeled; mean rise expected [E]
F02 Lubrication loss [E][I]	+1 to +6	0 to -0.3	0 to 0.3	1.1 to 1.6	drift, or step after service error	Whole cycle, worse when cold	0 to +0.05	0	+0.5 to +2
F03 Brake drag [E][I]	+1 to +4	0	0	1.0 to 1.2	step or intermittent	Whole cycle, present at idle	0 to +0.03	0	+1 to +4
F04 Bearing degradation [E][I]	+0.5 to +3	0	0 to 0.4	1.2 to 2.0	drift	Whole cycle, intermittent spikes 2 to 5 Nm	0 to +0.05	0	+0.3 to +1
F05 Backlash / looseness [E][I]	0 to +1	0	0.2 to 1.0 (broad)	1.1 to 1.5	drift	Spikes of 3 to 8 Nm at direction reversals (about 2.4 s and 6.0 s)	+0.05 to +0.3 (high for the dM)	0	0
F06 External drag [E][I]	+0.3 to +1	0	0	1.0 to 1.1	drift or step	Bump of 1 to 4 Nm at one phase window	0 to +0.02	0	0 to +0.5
F07 Overload / collision [E][I]	+3 to +12 (step) or spike	-2 to +3	0	1.0 to 1.5	step or single event	Spike at one phase for collision; whole cycle for payload	step +0.1 to +1.0	0 or step	0 to +3
F08 Drive/current fault [E][I]	+/-0.5 to +/-3 (offset)	0	non-harmonic lines (aliasing)	1.0 to 3.0	step or intermittent	Offset at all phases and at idle	0	0	same as offset
F09 Thermal [E][I]	+/-0.5 to +3	0	0	1.0	drift, reversible	Whole cycle; correlates with temperature	0	may be up	small
F10 Program/speed change [E]	-3 to +5	-3 to +3 or shape change	0	1.0	step at program boundary	Waveform shape changes	0 to small	0 or change	hold torque changes
F11 Sensor signal fault [E][I]	flat to 13.0, 0 or 45 (saturated)	-7 (flat)	0	0 (flat) or 5 (burst)	step	No cycle pattern, or constant	0	0	same
F12 Data-path fault [P][E]	apparent	apparent	unreliable	-	step	gaps, duplicates	other channels also affected	other channels also affected	-

3.1 Gearbox-wear consistency ratios

Because GEARBOX_WEAR is generated by fixed equations, its three effects are locked to one another through the severity $r = dM / 10.5$. This gives three quantitative tests that an agent can apply to a candidate window. A fault that raises dM but violates the ratios cannot be GEARBOX_WEAR as modeled [D].

Table 4. Consistency ratios for GEARBOX_WEAR [D]

Ratio	Formula	Expected value	Accept range for a 5-cycle window	Meaning if outside range
A6 per unit mean shift	$A6 / dM$	0.286 Nm/Nm	0.20 to 0.40 when dM is at least 2 Nm	Below 0.1: friction or load fault, not gear mesh ripple
Noise growth per unit mean shift	$(kN - 1) / dM$	0.143 per Nm	0.08 to 0.20	Below 0.03: offset or load change; above 0.3: bearing, looseness or sensor noise
TCP shift per unit mean shift	dT / dM	0.0524 mm per Nm	0.03 to 0.08 mm per Nm	Near 0: no kinematic effect; much larger: looseness or backlash
Swing change	dSw	0	within +/- 0.5 Nm	Nonzero: load or program change
Gun temperature	dG	0	within healthy band 37 to 58 degC	Change: thermal or process cause, not GEARBOX_WEAR in the repository

4. Detailed Failure Modes

4.1 F01 Gearbox wear: damaged gear mesh (GEARBOX_WEAR) [P]

Table 5. F01 fault profile

Attribute	Description
1. Fault name	GEARBOX_WEAR, gearbox wear (robot): damaged gear mesh in the axis 4 gearbox. Only fault scenario defined for robot-weld-01.
2. Affected component	Axis 4 gearbox (robot forearm, axis 4); secondary effect on wrist and tool flange path accuracy.
3. Physical mechanism	Tooth surface wear, pitting or tooth damage increases mesh friction and creates a periodic stiffness error. Each revolution of the damaged gear repeats a torque disturbance, so the drive must supply additional average torque plus a ripple at a fixed multiple of the cycle rate. Accumulating debris raises roughness.
4. Why Sensor 1 changes	The controller torque equals the demand needed to follow the programmed motion. Extra friction raises the mean demand (10.5 r Nm); the damaged mesh adds a 6x-cycle component (3.0 r Nm at 0.75 Hz); roughness raises noise by 1 + 1.5 r. Position error from the damaged mesh appears as TCP deviation (+0.55 r mm).
5. Expected telemetry pattern	Whole-cycle upward shift of the waveform (minimum rises from 4.25 to 14.72 Nm, peak from 21.75 to 32.28 Nm), unchanged 7.0 Nm swing, new 0.75 Hz line, wider noise; limit crossings only near the cycle peak (1.5 to 3.5 s). TCP mean rises with the same ramp.
6. Typical magnitude	dM 0 to +10.5 Nm; A6 0 to 3.0 Nm; noise 0.45 to 1.125 Nm; peak 21.75 to 32.28 Nm; warn at r 0.48 (M 18.0), alarm at r 0.82 (M 21.6).
7. Duration / persistence	Persistent once started and monotonic in the simulator (ramp then hold). Real wear is monotonic and irreversible without repair.
8. Rate of change	Simulator: 10.5 / rampSec Nm per second (0.0875 Nm/s at 120 s). Real world: days to months per 1 Nm [E][I].
9. Associated machine symptoms	Rising TCP deviation, audible roughness or growl at axis 4, weld position drift, increased servo load alarms, gearbox temperature rise [E]. TCP warns before torque (r 0.27 versus 0.48) [D].
10. Severity	Stage 1 to 2: low to medium; stage 3 (warn): high; stage 4 (alarm): very high; beyond model: critical.
11. Likely causes	Fatigue and overload history, lubricant contamination or degradation, high duty, collision history, wrong preload or mounting [E].
12. Alternative explanations	Lubrication loss (no ripple), brake drag (offset at idle), load change (swing changes), sensor offset (idle offset, no ripple).
13. Required inspection	Gearbox oil or grease sample for metal particles and colour; magnetic plug; backlash measurement at the flange; output-shaft play; noise and vibration check at low speed; TCP accuracy test; inspect for oil leakage [E].
14. Corrective action	Plan inspection at the next stop; do not run to alarm. Replace gearbox or gear set when damage is confirmed; flush and refill lubricant per OEM; recheck mastering and calibration; return to service with verification run.
15. Post-repair expected behaviour	Cycle mean returns to 13.0 +/- 0.5 Nm, A6 at most 0.2 Nm, noise 0.45 to 0.55 Nm, peak about 21.75 Nm, TCP mean about 0.10 mm; status normal.

4.2 F02 Lubrication degradation or loss [E]

Table 6. F02 fault profile

Attribute	Description
1. Fault name	Lubrication degradation (grease starvation, oxidised or contaminated grease, wrong grade, low oil level) in the axis 4 gearbox.
2. Affected component	Axis 4 gearbox and its bearings: every lubricated contact (gear mesh, output bearing, seals). Secondary effect on the motor through higher load.
3. Physical mechanism	Without an adequate lubricant film the contacts move from fluid to mixed or boundary lubrication. Viscous drag rises when the lubricant is too thick (cold, oxidised), Coulomb friction rises when the film is thin, and frictional heating accelerates oxidation. Left uncorrected, asperity contact starts abrasive wear and the condition develops into F01.
4. Why Sensor 1 changes	The drive must overcome the extra friction for the same programmed motion, so the demanded torque rises by roughly a constant amount over the cycle: dM positive, swing unchanged. Because friction does not create a periodic disturbance tied to a gear tooth, there is no 6x line. Roughness of mixed lubrication raises noise modestly.
5. Expected telemetry pattern	Whole-cycle upward offset with the 7.0 Nm swing preserved; A6 below 0.4 Nm; noise up 10 to 60 percent; the offset is largest right after a cold start or long stop and decays over minutes as the gearbox warms; idle holding torque slightly raised.
6. Typical magnitude	dM +1 to +6 Nm (S equivalent 0.1 to 0.57 but without ripple); kN 1.1 to 1.6; A6 at most 0.3 Nm; TCP change 0 to +0.05 mm [I]. Peak can reach the warn limit at dM above about 4.3 Nm, so a warn status without ripple is a lubrication or friction indicator.
7. Duration / persistence	Persistent, with a reversible warm-up component of about 0.5 to 1.5 Nm in the first minutes; the persistent part is removed only by service.
8. Rate of change	Slow drift of M of 0.02 to 0.5 Nm per day as grease ages; a step of 1 to 4 Nm if a wrong lubricant or an under-fill was introduced at maintenance [E][I].
9. Associated machine symptoms	Warm gearbox housing, grease discolouration or bleed, seal weeping, rising idle torque, servo load warnings at start-up, longer warm-up behaviour. No effect on weld gun temperature.
10. Severity	S1 to S3 depending on dM. Becomes S4 if friction reaches the alarm level, and it is a precursor of F01, so it is treated one level higher when the lubrication interval is overdue.
11. Likely causes	Missed relubrication, seal failure or leakage, overheating (oxidation), wrong grease grade or mixing incompatible greases, water or spatter ingress, over-filling producing churning losses.
12. Alternative explanations	F01 (check whether A6 / dM is above 0.2); F03 brake drag (idle offset larger, brake area warm); F09 thermal (reversible and correlated with temperature); F10 payload (swing changes).
13. Required inspection	Lubricant level, colour, smell and particle content; seals and leak paths; gearbox surface temperature at steady state; maintenance record of last lubrication and grade; compare cold-start versus warm torque.
14. Corrective action	Drain and refill or relubricate per OEM grade and quantity; replace seals; clean breather; run a warm-up program and re-baseline; schedule a follow-up trend check after 5 days of operation. If metal particles are found, switch the working diagnosis to F01.
15. Post-repair expected behaviour	After service and warm-up, M returns to 13.0 +/- 0.5 Nm, noise to 0.45 to 0.55 Nm, A6 stays at most 0.2 Nm, warm-up offset at most 0.5 Nm, platform status normal.

4.3 F03 Brake drag (partially engaged holding brake) [E]

Table 7. F03 fault profile

Attribute	Description
1. Fault name	Brake drag: the axis 4 motor holding brake does not release fully, releases late, or re-engages lightly during motion.
2. Affected component	Axis 4 motor brake (friction disc, armature, coil) and its supply circuit; secondary heating of the motor and gearbox.
3. Physical mechanism	A friction disc held lightly against the rotor by marginal release voltage, worn parts or a stuck armature applies a speed-independent Coulomb torque opposing motion. The motor must supply this torque continuously, and generates heat in the brake and motor.
4. Why Sensor 1 changes	Extra constant torque in the direction of motion appears as a positive offset in the demanded torque during motion, and a non-zero holding offset at idle. The cycle swing is unchanged and no periodic mesh disturbance exists, so there is no 6x line.
5. Expected telemetry pattern	Cycle shape preserved, shifted up by a near-constant amount; A6 and noise unchanged; torque during idle or hold also elevated; may appear only after the brake has been energised for a while (intermittent); brake area warm.
6. Typical magnitude	dM +1 to +4 Nm; idle offset +1 to +4 Nm; kN 1.0 to 1.2; TCP 0 to +0.03 mm [I].
7. Duration / persistence	Persistent if mechanical; intermittent if linked to supply voltage or temperature.
8. Rate of change	Step at onset (for example after a brake replacement or after a power event), not a gradual ramp [E].
9. Associated machine symptoms	Brake release click absent or late, motor and brake housing warm, drive current high at standstill, following error at motion start, possible brake alarm in controller.
10. Severity	S2 to S3; heat damage is possible if not corrected, so it is escalated if idle offset exceeds 3 Nm.
11. Likely causes	Low brake supply voltage, worn friction disc, contamination or corrosion of the armature, incorrect release timing in controller, brake lining swelling.
12. Alternative explanations	F02 lubrication (idle offset smaller, no brake heat); F08 drive offset (the offset persists with the brake mechanically free); F11 sensor offset.
13. Required inspection	Brake supply voltage at the brake terminals during release; release and engage timing; armature air gap; torque required to turn the axis by hand with servo off and brake released (per OEM procedure); brake temperature.
14. Corrective action	Correct supply voltage and wiring; adjust air gap; replace disc or brake assembly; correct release timing; verify no residual drag.
15. Post-repair expected behaviour	Idle offset removed (idle torque at baseline hold value), M returns to 13.0 +/- 0.5 Nm, brake area at normal temperature.

4.4 F04 Bearing degradation (motor, gearbox output, forearm support) [E]

Table 8. F04 fault profile

Attribute	Description
1. Fault name	Bearing degradation in the axis 4 motor, gearbox output stage or forearm support bearings.
2. Affected component	Axis 4 bearings, with possible secondary damage to shaft, seals and gear mesh.
3. Physical mechanism	Spalling of a race or rolling element, contamination or lubricant starvation increases rolling resistance and produces repeated impacts when a defect passes a rolling element. Defect frequencies are tens to hundreds of Hz for normal speeds.
4. Why Sensor 1 changes	Extra rolling resistance slightly raises the mean demand. Impacts perturb the torque estimate as short irregular spikes and broadband noise. The defect frequencies are far above the 1 Hz observable limit of Sensor 1 at 2 Hz sampling, so only the broadband effect and spike count are visible, not a harmonic line.
5. Expected telemetry pattern	Noise ratio 1.2 to 2.0; sparse spikes of 2 to 5 Nm not locked to one phase; small mean rise; A6 below 0.4 Nm; effect grows over weeks.
6. Typical magnitude	dM +0.5 to +3 Nm; kN 1.2 to 2.0; spike amplitude 2 to 5 Nm [I].
7. Duration / persistence	Persistent with intermittent spikes, worsening.
8. Rate of change	Drift over weeks to months, accelerating near end of life [E].
9. Associated machine symptoms	Audible rumble, local temperature rise at the housing, play at the flange, metallic particles in grease.
10. Severity	S2 to S4; high near end of life because seizure can follow.
11. Likely causes	Contamination, over-torquing, misalignment, electrical discharge through the bearing, lubricant loss (F02), overload (F07).
12. Alternative explanations	F01 (ripple at 0.75 Hz present in F01 and absent in F04); F05 (spikes locked to reversals); F11 burst noise (no mean effect, not persistent).
13. Required inspection	Axial and radial play by hand or indicator; bearing temperature; grease analysis; vibration measurement with a separate high-rate accelerometer (not available in Sensor 1).
14. Corrective action	Replace the bearing set, inspect shaft and seats, relubricate, correct alignment; re-baseline after repair.
15. Post-repair expected behaviour	Noise ratio back to 1.0 to 1.2, spikes eliminated, M within 0.5 Nm of baseline.

4.5 F05 Backlash growth or mechanical looseness [E]

Table 9. F05 fault profile

Attribute	Description
1. Fault name	Backlash growth or mechanical looseness: excess gear backlash, worn or loose coupling, loose mounting bolts of the axis 4 interface.
2. Affected component	Gearbox-to-arm interface, motor coupling or key, mounting bolts and dowels.
3. Physical mechanism	Free play lets the load travel before the drive engages. When the sign of the load torque reverses, the drive crosses the lash gap and the contact re-engages with an impact. The impact is seen as a torque spike and the motion error as TCP scatter.
4. Why Sensor 1 changes	At each torque reversal the controller applies a short burst of corrective torque; the spike is repeated every cycle at the same phase. Mean and swing change little because free play does not add continuous friction.
5. Expected telemetry pattern	Short spikes of 3 to 8 Nm at the reversal phases of the cycle (near 2.4 s where torque peaks, and near 6.0 s where it is at its minimum); small broadband harmonics 2x to 8x of the cycle of 0.2 to 1.0 Nm; TCP deviation rises more than the torque would suggest.
6. Typical magnitude	dM 0 to +1 Nm; spikes 3 to 8 Nm; A6 0.2 to 1.0 Nm broadband; dT +0.05 to +0.3 mm [I].
7. Duration / persistence	Persistent; spikes repeat each cycle.
8. Rate of change	Drift over weeks (wear) or a step when a fastener loosens [E].
9. Associated machine symptoms	Clunk at reversal, repeatability loss, weld position scatter, loose bolts or fretting marks.
10. Severity	S2 to S4; high if fasteners can fail or coupling can break.
11. Likely causes	Loosened fasteners (vibration, thermal cycling), coupling wear, gear wear (can progress from F01), impact or collision history.
12. Alternative explanations	F01 (in F01 the mean rises by 10.5 r and the ripple is 6x, not broadband spikes); F04 (spikes not locked to reversals); F06 (bump rather than spike).
13. Required inspection	Fastener torque check; backlash measurement at the flange with a dial indicator; coupling and key condition; fretting marks; compare TCP repeatability.
14. Corrective action	Retorque to specification and apply thread-locking per OEM; replace coupling or key; adjust or replace gear set if backlash is out of limit; re-master axis.
15. Post-repair expected behaviour	Spikes at the reversals disappear, broadband harmonics return to baseline, TCP mean back to about 0.10 mm.

4.6 F06 External drag: cable dress pack, hoses and seals [E]

Table 10. F06 fault profile

Attribute	Description
1. Fault name	External drag from the cable dress pack, hoses or spatter build-up around seals and covers on axis 4.
2. Affected component	Dress pack, hose guide, covers, spatter shields and seals around axis 4.
3. Physical mechanism	Cables and hoses are tensioned or pinched when the forearm reaches a certain posture, adding a resisting torque. Spatter accumulation adds friction at covers and seals.
4. Why Sensor 1 changes	The extra torque is posture dependent, so it appears only when the arm passes through the affected stroke segment: a bump at a fixed phase of every cycle.
5. Expected telemetry pattern	Bump of 1 to 4 Nm centred on one phase window each cycle, width 0.5 to 1.5 s; the rest of the cycle unchanged; small mean rise; A6 unchanged; position dependent so it moves if the program path changes.
6. Typical magnitude	dM +0.3 to +1.0 Nm; bump 1 to 4 Nm; kN 1.0 to 1.1 [I].
7. Duration / persistence	Persistent; position dependent.
8. Rate of change	Step after a tool or cable change; slow rise with spatter [E].
9. Associated machine symptoms	Visible cable wear, tight dress pack, jerky motion at one posture.
10. Severity	S1 to S2; cable damage risk if tension is high.
11. Likely causes	Incorrect routing or slack after tool change, worn guides, spatter build-up.
12. Alternative explanations	F05 (impulsive spike, not a smooth bump); F07 collision (single event); F01 (affects the whole cycle).
13. Required inspection	Dress pack routing and slack across the full stroke; free movement by hand of hoses; spatter on covers and seals.
14. Corrective action	Re-route and adjust slack; replace worn guides; clean spatter; fit protective covers.
15. Post-repair expected behaviour	Bump removed; template returns to the healthy shape; M within 0.5 Nm of baseline.

4.7 F07 Overload, payload increase or collision [E]

Table 11. F07 fault profile

Attribute	Description
1. Fault name	Overload: payload increase, wrong tool mass data or a collision of arm, tool or gun with an obstacle.
2. Affected component	Forearm structure, wrist, tool flange and gun, with gearbox and bearings at risk after a collision.
3. Physical mechanism	Added mass requires more torque for the same motion in proportion to mass; a collision adds an impulsive reaction torque and may damage gear teeth or bearings.
4. Why Sensor 1 changes	A heavier payload raises the mean and the swing together; a collision causes one or a few very high samples followed by recovery, and possible damage afterwards.
5. Expected telemetry pattern	Payload: step in M and in swing, same shape, no ripple. Collision: spike up to the clamp of 45 Nm, a step in TCP, possible persistent offset and later development of F01 or F05.
6. Typical magnitude	Payload: dM +3 to +12 Nm and swing +1 to +3 Nm. Collision: spike 30 to 45 Nm [I].
7. Duration / persistence	Payload persistent until removed; collision single event with possible persisting damage.
8. Rate of change	Step within one cycle [E].
9. Associated machine symptoms	Collision stop, safety controller event, damaged tool or gun, path offset.
10. Severity	S3 to S5; collision escalates to S5 until inspected.
11. Likely causes	Teaching error, obstruction in the cell, wrong tool data, process change.
12. Alternative explanations	F01 (gradual and with ripple), F10 (program change, no spike), F11 (spike with no TCP effect).
13. Required inspection	Visual damage, tool mass data, backlash and play after collision, mastering position, gearbox noise.
14. Corrective action	Remove cause, correct tool data, inspect gear train and bearings after any collision, re-master the axis, run a verification cycle.
15. Post-repair expected behaviour	Mean and swing return to baseline, spike gone, TCP back to about 0.10 mm.

4.8 F08 Drive or current-measurement fault [E]

Table 12. F08 fault profile

Attribute	Description
1. Fault name	Drive or current-measurement fault: offset, gain error or phase imbalance in the axis 4 servo drive or its current sensors.
2. Affected component	Axis 4 servo drive, motor current sensors, motor cable and encoder interface.
3. Physical mechanism	A current sensor offset or gain error makes the controller's torque estimate differ from the true torque. Phase imbalance produces torque ripple at electrical frequencies, which are far above Nyquist and alias into the observable band.
4. Why Sensor 1 changes	The reported torque is offset or scaled relative to the physical torque, while the mechanics are unaffected. Aliased ripple appears as lines at frequencies that are not multiples of 0.125 Hz.
5. Expected telemetry pattern	Constant offset present at all phases and at idle; swing scaled if the gain is wrong; non-harmonic spectral lines; no TCP change because the mechanics are fine.
6. Typical magnitude	dM +/- 0.5 to 3 Nm; swing scale error 2 to 10 percent; non-harmonic lines [I].
7. Duration / persistence	Persistent or temperature dependent.
8. Rate of change	Step [E].
9. Associated machine symptoms	Drive diagnostics, encoder or current imbalance warnings, drive overtemperature.
10. Severity	S2 to S3 (misleading data and possible drive damage).
11. Likely causes	Failed current sensor, cable fault, ageing of drive electronics, loose connector.
12. Alternative explanations	F03 brake drag (offset with warm brake); F11 sensor offset in the gateway; F01 (check ripple and TCP).
13. Required inspection	Drive diagnostic log, phase current balance, motor cable and connectors, encoder signal, comparison with another torque estimate if available.
14. Corrective action	Replace sensor, drive module or cable; recalibrate current measurement; verify gain with a known load.
15. Post-repair expected behaviour	Idle offset at zero, non-harmonic lines absent, M within 0.5 Nm of baseline.

4.9 F09 Thermal effect: motor or gearbox temperature [E]

Table 13. F09 fault profile

Attribute	Description
1. Fault name	Thermal effect: motor or gearbox overtemperature, or cold-start viscosity.
2. Affected component	Axis 4 motor windings, magnets, gearbox lubricant and housing.
3. Physical mechanism	Lubricant viscosity falls with temperature and rises when cold; the motor torque constant falls with magnet temperature, so more current is needed for the same torque.
4. Why Sensor 1 changes	Friction demand and the current-to-torque relation depend on temperature, so M drifts reversibly with temperature without changing the cycle shape.
5. Expected telemetry pattern	Slow reversible drift of M correlated with temperature or with time after start; unchanged shape; A6 not present; no TCP change.
6. Typical magnitude	dM +/- 0.5 to +3 Nm [I].
7. Duration / persistence	Reversible; follows thermal time constants.
8. Rate of change	Minutes to hours [E].
9. Associated machine symptoms	Overtemperature warnings, hot housings, long production without pause.
10. Severity	S1 to S2; S3 if overtemperature protection trips.
11. Likely causes	Cold start, hot ambient, blocked cooling, high duty.
12. Alternative explanations	F02 (does not reverse with temperature); F08 (offset is independent of temperature).
13. Required inspection	Temperature at motor and gearbox; cooling fans and filters; ambient; duty cycle.
14. Corrective action	Allow warm-up; restore cooling; reduce duty; adjust ambient.
15. Post-repair expected behaviour	M returns to baseline at equilibrium temperature.

4.10 F10 Control-system, program or tuning change [E]

Table 14. F10 fault profile

Attribute	Description
1. Fault name	Control-system change: new program, speed override, payload data or servo gain change.
2. Affected component	Robot controller, motion program, servo parameters.
3. Physical mechanism	Motion profile and gains determine accelerations and torque noise; they are changes, not defects.
4. Why Sensor 1 changes	Demanded torque follows the program, so the waveform shape, mean and swing change.
5. Expected telemetry pattern	Step or shape change at a program boundary; swing and mean change; no 0.75 Hz line; the weld cycle length may change; TCP and gun temperature may also change.
6. Typical magnitude	dM -3 to +5 Nm; swing -3 to +3 Nm [I].
7. Duration / persistence	Persistent until the program changes back.
8. Rate of change	Step [E].
9. Associated machine symptoms	Program ID change, operator action, speed override.
10. Severity	None; it is a non-fault condition that must not be reported as wear.
11. Likely causes	Planned change, new product, new tool.
12. Alternative explanations	F07 payload (both change mean and swing).
13. Required inspection	Program ID, speed override, payload data, controller change log.
14. Corrective action	None; re-baseline for the new program.
15. Post-repair expected behaviour	A new baseline is established for the new program.

4.11 F11 Sensor signal fault: flatline, saturation, offset or spike train [E]

Table 15. F11 fault profile

Attribute	Description
1. Fault name	Signal fault in the torque estimate path: flatline, saturation at 0 or 45 Nm, offset or burst noise.
2. Affected component	Torque estimate path in the controller interface or gateway firmware.
3. Physical mechanism	A stuck value, scaling error, firmware fault or interference makes the reported value no longer follow the axis.
4. Why Sensor 1 changes	The reported value no longer follows the axis.
5. Expected telemetry pattern	Flat constant value with zero variance, values clipped at 0 or 45 Nm, offset without TCP change, or burst noise unrelated to the cycle.
6. Typical magnitude	Constant value; variance 0; clipping at 0 or 45; noise ratio 5 and above [I].
7. Duration / persistence	Persistent or intermittent.
8. Rate of change	Step.
9. Associated machine symptoms	Other channels normal; diag shows heap or reconnect anomalies; twin shows constant colour.
10. Severity	S2 to S3 (diagnostic blind spot).
11. Likely causes	Firmware bug, scaling error, wiring, interference, frozen task.
12. Alternative explanations	F12 data-path fault; idle robot (a flat trace with the program stopped).
13. Required inspection	Raw gateway log, diag fields, other channels, program state.
14. Corrective action	Fix firmware or wiring; restart gateway; verify scaling.
15. Post-repair expected behaviour	Cycle pattern returns; variance consistent with 0.45 Nm noise.

4.12 F12 Data-path fault: stale, sequence gaps, clock error, identity conflict [P]

Table 16. F12 fault profile

Attribute	Description
1. Fault name	Data-path fault: stale data, packet loss, reboot, clock not synced or identity conflict.
2. Affected component	ESP32 gateway, Wi-Fi link, MQTT broker and backend ingestion.
3. Physical mechanism	Link, clock or identity problems affect the telemetry, not the axis.
4. Why Sensor 1 changes	Samples are missing, delayed, duplicated or mis-stamped, so features computed from them are wrong.
5. Expected telemetry pattern	Gaps in seq, device stale after max(5 s, 4 x interval), status offline about 15 s after power loss, synced false, or two devices claiming one machine ID.
6. Typical magnitude	Missing samples; jitter; stamp errors [P].
7. Duration / persistence	Persistent or intermittent.
8. Rate of change	Step.
9. Associated machine symptoms	All three channels affected together; diag shows wifiReconnects or mqttReconnects.
10. Severity	Low for the machine, high for diagnosis.
11. Likely causes	Wi-Fi quality (5 GHz only, weak signal), broker credentials, blocked NTP, simulator and board both running.
12. Alternative explanations	F11 signal fault (only one channel affected).
13. Required inspection	Gateway diag fields, broker connection, clock sync, device list for identity conflict.
14. Corrective action	Fix network, clock or identity.
15. Post-repair expected behaviour	No gaps; synced true; one device per machine ID.

5. Diagnostic Reasoning: Anatomy of the GEARBOX_WEAR Signature

This section explains why each element of the F01 signature exists and in which order the elements become detectable. The agent should use it to justify a diagnosis and to judge how early a developing fault can be recognised.

5.1 Why each component of the signature exists

Table 17. Term-by-term interpretation of the GEARBOX_WEAR torque equation [P]

Term in equation	Magnitude at severity r	Physical reading	Diagnostic consequence
+10.5 r	0 to 10.5 Nm	Added friction and load from a damaged mesh: the drive supplies extra mean torque all cycle	Raises minimum, mean and peak equally; this is the earliest broad evidence
+3.0 r sin(12 pi p)	0 to 3.0 Nm at 0.75 Hz	Periodic disturbance repeating six times per cycle from the damaged gear mesh	Most specific evidence; absent from friction-only faults
0.45 (1 + 1.5 r) x Gaussian	0.45 to 1.125 Nm	Roughness of the contact adds random torque variation	Supporting evidence; also raised by bearing faults and sensor noise
Unchanged 7.0 sin(2 pi p) and 2.2 sin(8 pi p)	7.0 and 2.2 Nm	Programmed motion does not change	Excludes load and program changes
Companion: +0.55 r + 0.05 r sin(2 pi t / 17) mm in TCP deviation	0 to 0.55 mm	Kinematic error from a damaged gear train reaches the tool	Corroboration; wear is visible at the tool earlier than torque limits

5.2 Order of detectability over the default 120 s ramp

Table 18. Detection lead times for GEARBOX_WEAR at rampSec = 120 s [D]

Evidence	Threshold	Reached at severity r	Time (s)	Platform status at that time	Lead over first torque warn (s)
Cycle mean M above baseline band	13.5 Nm	0.048	5.7	normal	51.9
Sixth harmonic A6 above watch level	0.4 Nm	0.133	16.0	normal	41.6
Noise sigma above 0.55 Nm	0.55 Nm	0.148	17.8	normal	39.8
Cycle mean M reaches early-degradation boundary	15.0 Nm	0.190	22.8	normal	34.8
TCP deviation mean reaches warnHigh	0.25 mm	0.265	31.8	normal for torque, warn for TCP	25.8
First torque warn peak	26 Nm	0.48	57.6	warn	0
TCP deviation mean reaches alarmHigh	0.5 mm	0.72	86.4	alarm for TCP	-28.8
First torque alarm peak	30 Nm	0.82	98.4	alarm	-40.8
Full severity plateau	32.3 Nm peak	1.0	120	alarm	-62.4

The planned evaluation of the diagnostic agent includes detection lead time against a threshold-alarm baseline [P]. The table shows what is available: feature-based detection gives about 40 to 50 s of lead at the default ramp (about one third of the ramp), and the TCP channel warns about 26 s before the torque limit is reached.

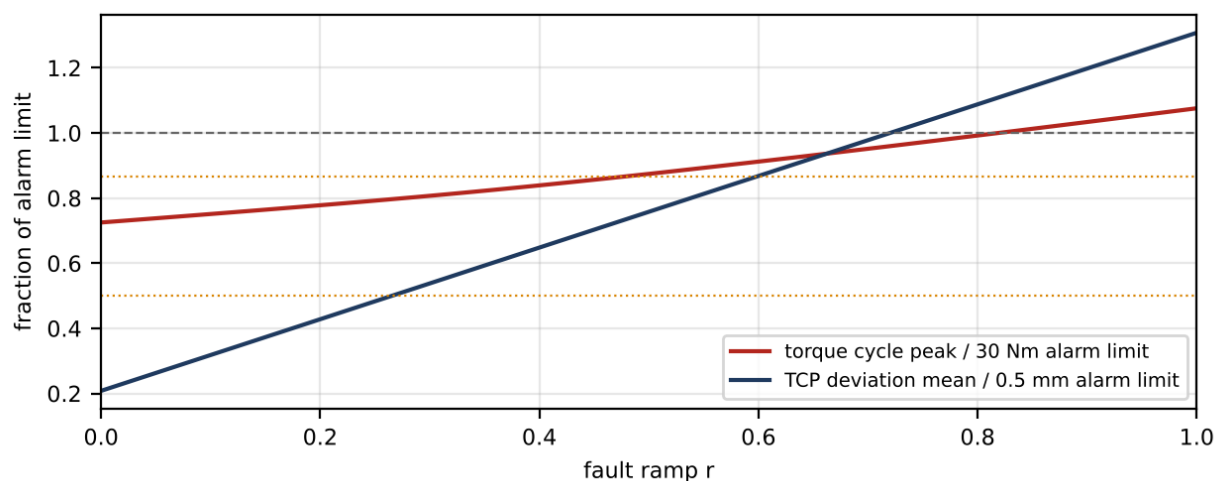


Figure 3. Fraction of the alarm limit reached by Sensor 1 cycle peak and by TCP deviation mean as GEARBOX_WEAR progresses: TCP leads torque [D].

5.3 Where in the cycle the limits are crossed

Table 19. Time window in the 8 s cycle with torque at or above the limit (noise-free) [D]

Severity r	Time above 26 Nm (first to last)	Fraction of cycle above 26 Nm	Time above 30 Nm (first to last)	Fraction of cycle above 30 Nm
0.50	2.55 to 2.82 s	3.4 percent	none	0 percent
0.60	1.68 to 3.00 s	16.6 percent	none	0 percent
0.75	0.26 to 3.14 s	27.0 percent	none	0 percent
0.90	0.15 to 3.24 s	32.9 percent	1.67 to 3.01 s	7.8 percent
1.00	0.10 to 4.43 s	39.5 percent	0.31 to 3.10 s	17.8 percent

IF the samples at or above 26 Nm occur between about 0 and 4.4 s of the cycle and none occur in the weld window THEN the pattern is consistent with a raised baseline (F01, F02, F03, F07 payload); IF the limit is crossed only inside a narrow phase window of less than 1 s at an unusual phase THEN suspect a local event (F05, F06, F07 collision) rather than a baseline shift.

5.4 Phase map of the cycle

Table 20. Phase map for interpreting phase-locked patterns [D][E]

Time in cycle (s)	Healthy torque (Nm, noise-free)	Arm state (engineering reading) [E]	Faults that show most here
0.0 to 1.0	13.0 to 17.9 (rising)	Accelerating into the stroke	F06 drag bump (example 1.2 s), F03, F02
1.0 to 2.4	17.9 to 21.75 (peak)	Approach of maximum gravity and acceleration load	F01 lobes, F07 payload, F05 reversal near 2.4 s
2.4 to 4.4	21.75 to 13.0 (falling)	Deceleration, approach to weld pose	F05 reversal spikes, F01 lobes
4.4 to 6.0	about 13 to 4.2 (weld window, gun on)	Holding pose for spot welding	Weld gun reaction, F06, F08 offset, F03 offset
6.0 to 8.0	4.25 to 13.0 (rising)	Return motion	F05 reversal near 6.0 s, F06

6. Fault Progression and Transition Paths

Table 21. How one fault can lead to another

From	To	Mechanism	Evidence of transition in Sensor 1	Action to stop the transition
F02 lubrication loss	F01 gearbox wear	Boundary contact leads to abrasive wear and pitting	A6 appears and grows above 0.4 Nm on top of a friction offset	Relubricate early; check oil for particles
F07 collision	F01, F05, F04	Tooth or bearing damage, loosened fasteners	Step in M at the event, later ripple or spikes	Inspect after any collision
F05 backlash	F01 gearbox wear	Impacts accelerate tooth wear	Spikes at reversals followed by rising mean and ripple	Correct backlash and fasteners
F06 external drag	F02 (spatter into seals)	Seal damage leads to lubricant loss	Bump followed by uniform offset	Clean and replace seals
F03 brake drag	F09 thermal / F02	Heat degrades grease	Idle offset then thermal drift	Repair brake
F04 bearing	F01, F05	Bearing play changes mesh and backlash	Noise rise then ripple	Replace bearing
F08 drive offset	None (data fault)	Masks real condition	Offset present at idle	Recalibrate

6.1 Pairs of faults that are easy to confuse

Table 22. Ambiguity table for look-alike patterns

Pair	Why they look alike	Minimum extra evidence to separate	Decisive test
F01 versus F02	Both raise M and may reach warn	A6 / dM and (kN - 1) / dM	Ratio test; oil particle analysis
F02 versus F09	Both raise M without ripple	Correlation with temperature, reversibility	Compare cold start and warm state
F03 versus F08	Both give offset at idle	Brake temperature and release timing	Brake voltage and drive log
F05 versus F04	Both give spikes	Phase locking of spikes to reversals	Phase histogram of spikes
F07 versus F10	Both change mean and swing	Program ID, event log	Program log
F11 versus F12	Both give no usable signal	Whether other channels are also affected	Check seq, status and diag
F01 versus F05 early	Both raise TCP and add low harmonics	6x line versus broadband spikes	Spectral shape and reversal spikes

7. Threshold-Based Fault Differentiation

The table below separates states using the cycle mean shift dM (equivalently $S = dM / 10.5$) so that the agent can state which fault families are plausible at each magnitude. The first two rows use the project limits; the rest are proposals [I].

Table 23. Threshold-based differentiation by magnitude of mean shift and companion features

Band	dM (Nm)	M (Nm)	Peak per cycle (Nm)	Platform status	Plausible fault families	Discriminator to apply
Normal	-0.5 to +0.5	12.5 to 13.5	21.75, max 23.1	normal	none	none
Mild deviation	+0.5 to +2.0	13.5 to 15.0	22.3 to 23.7	normal	F01 stage 1, F02, F03, F06, F09, F10	A6 above 0.4, idle offset, phase-locked bump, temperature
Early degradation	+2.0 to +5.0	15.0 to 18.0	23.7 to 26.0	normal (rare warn)	F01 stage 2, F02, F03, F07 payload, F08	Consistency ratios, swing change, idle offset
Warning	+5.0 to +8.6	18.0 to 21.6	26.0 to 30.0	warn	F01 stage 3, F02 severe, F07, F03 severe	A6/dM about 0.29; TCP rise
Severe abnormality (alarm)	+8.6 to +10.5	21.6 to 23.5	30.0 to 32.3	alarm	F01 stage 4, F07	TCP above 0.5 mm supports F01
Critical	above +10.5 or sample at or above 34	above 23.5	above 32.3	alarm	F07 collision, F01 beyond model, F11 saturation	Event log, step vs ramp

7.1 Sensor pattern to likely fault

Table 24. Sensor pattern to most likely fault

Observed Sensor 1 pattern	First candidate	Second candidate	Third candidate	Key test
Mean up, A6 up, noise up in the consistency ratios (Section 3.1), TCP up	F01 gearbox wear	F05 looseness	F04 bearing	Ratios (Section 3.1)
Mean up, A6 about 0, noise slightly up, swing unchanged	F02 lubrication	F09 thermal	F03 brake drag	Warm-up behaviour, idle torque
Mean up and idle torque up	F03 brake drag	F08 drive offset	F02 lubrication	Idle offset size and brake release
Mean and swing both up proportionally	F07 payload	F10 program change	-	Program ID, payload data
Single high spike then normal	F07 collision	F11 spike	F12 data glitch	Repeat at same phase, TCP step
Spikes at fixed phases each cycle	F05 backlash	F04 bearing	F06 external drag	Phase of reversals vs bump
Bump at one phase window	F06 external drag	F05	F01 early	Width, location, other phases
Flat constant value	F11 flatline	F12 stale	Idle state	Variance, status, other channels
Values at 0 or 45	F11 saturation	F08	-	Clamp behaviour
Gaps, duplicates	F12	-	-	seq, bootId, synced
Everything normal but device flagged	False alarm	F12	-	Per-sample status vs per-cycle features

7.2 Discriminator matrix

Table 25. Which fault explains which feature (Y = expected, N = not expected, P = possible)

Feature	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11
Mean rise	Y	Y	Y	P	N	P	Y	P	P	P	N
A6 ripple above 0.4	Y	N	N	P	P	N	N	N	N	N	N
Noise rise	Y	P	N	Y	P	N	N	P	N	N	P
Swing change	N	N	N	N	N	N	P	N	N	P	N
Idle offset	N	P	Y	P	N	N	N	Y	P	P	Y
Phase-locked bump	N	N	N	N	P	Y	P	N	N	P	N
Spikes	N	N	N	P	Y	N	Y	N	N	N	P
TCP rise	Y	P	N	P	Y	N	P	N	N	P	N
Gun temperature change	N	N	N	N	N	N	N	N	P	P	N
Reversible with temperature	N	P	N	N	N	N	N	N	Y	N	N

8. Symptom-to-Cause Mapping

Table 26. Machine symptom to possible cause

Symptom	Sensor 1 pattern	Possible causes (ranked)	Inspection first	Basis
Weld spots drift off position	Mean up, TCP up	F01, F05, F07	Backlash and TCP test	[E]
Warning on twin	Peak 26 to 30 Nm	F01, F02, F07	Trend of M and A6	[P]
Alarm on twin	Peak 30 Nm or above	F01, F07	Stop planning, inspect gearbox	[P]
Axis 4 growl or noise	Noise up, A6	F01, F04	Gear and bearing inspection	[E]
Axis hot at stop	Mean up, idle offset	F03, F02, F09	Brake and lubricant	[E]
Clunk at reversal	Spikes at fixed phases	F05	Fasteners and backlash	[E]
After tool change	Mean/swing step	F10, F07	Tool data	[E]
After collision	Spike then step	F07	Mechanical damage	[E]
Gun hot but torque normal	None	Process cause	Gun channel	[P]
No data	Stale	F12	Gateway	[P]
Flat value	No cycle	F11, F12	Signal path	[E]

9. Severity Classification

Table 27. Severity classification for Sensor 1 conditions

Seve rity	Name	Sensor 1 criteria	Platform status	Response time	Consequence if ignored
S0	Normal	All features healthy	normal	none	-
S1	Advisory	M 13.5 to 15.0 Nm, A6 0.4 to 0.6 Nm	normal	review within shift	Continued wear
S2	Minor	M 15.0 to 18.0 Nm	normal or occasional warn	plan inspection within days	Warn begins
S3	Major	Peak at or above 26 Nm for 2 cycles	warn	inspect at next stop	Alarm and downtime
S4	Severe	Peak at or above 30 Nm for 2 cycles	alarm	intervene within shift	Gearbox failure
S5	Critical	Sample at or above 34 Nm or M at or above 23.5	alarm	immediate	Failure, safety risk

10. Alternative Explanations and Inspection Requirements

Table 28. Alternative explanations and how to exclude them

Primary hypothesis	Alternative explanation	Evidence that would favour the alternative	How to exclude
F01 gearbox wear	F02 lubrication loss	Mean up without A6; recovery with warm-up	Check A6 ratio; temperature trend
F01 gearbox wear	F03 brake drag	Idle torque offset	Compare idle torque
F01 gearbox wear	F07 payload	Swing change	Check swing and program ID
F01 gearbox wear	F08 offset	No TCP change, offset at idle	TCP and idle
F01 gearbox wear	F10 program change	Step at program boundary, no ripple	Program log
F07 collision	F11 data glitch	TCP unchanged, single sample	Next cycle and other channels
Flatline sensor	Idle robot	Status online but program stopped	Program state
Alarm status	False alarm from flicker	Per-cycle peak below 26	Cycle features

Table 29. Inspection requirements by fault

Fault	Inspection	Tools	Requires stop?	Pass criterion [I]
F01	Oil sample, backlash, noise, TCP test	Particle test, dial indicator	Yes, at planned stop	Backlash within OEM, no metal particles, TCP at 0.10 mm
F02	Lubricant level and condition	Visual, sample kit	Yes	Grease per OEM
F03	Brake release voltage and timing	Multimeter	Yes	Supply within OEM
F04	Bearing play and noise	Stethoscope, indicator	Yes	Play within OEM
F05	Fastener torque, backlash	Torque wrench	Yes	Torque per OEM
F06	Dress pack	Visual	Short stop	No drag
F07	Mechanical damage, tool data	Visual, program	Yes	No damage
F08	Drive diagnostics	Diagnostic software	Yes	No fault
F09	Temperature	Thermometer	No	In range
F11, F12	Signal path and gateway	diag fields, logs	No	Clean data

11. Summary, Safety Notes, References and Revision

11.1 Diagnostic summary

- Only the combination of mean rise, 0.75 Hz ripple and noise increase in the proportions of the consistency ratios (Section 3.1), together with a TCP rise and unchanged gun temperature, is the signature of GEARBOX_WEAR as modeled [P][D].
- Friction-type faults (F02, F03, F09) raise the mean without ripple; mechanical event faults (F05, F06, F07) are phase-locked or single; electrical and sensor faults produce offsets, flat traces or non-harmonic lines; data-path faults affect every channel.
- Key thresholds: warn 26 Nm, alarm 30 Nm [P], critical 34 Nm [I], mean shift 1.0 Nm watch, A6 0.4 Nm watch.

11.2 Safety and escalation notes

A collision or critical condition requires a safe stop and inspection by a qualified person before restart. All inspection work on the axis requires lock-out and tag-out and the OEM procedure; this document does not replace the OEM manual or site safety rules.

11.3 References and engineering basis

- Repository simulation equations for GEARBOX_WEAR and the MQTT contract fault table [P].
- General engineering knowledge of robot servo gearbox diagnostics, current signature analysis and condition monitoring practice [E]; no claim of OEM validation.
- Illustrative magnitudes are tagged [I] and must be recalibrated with real data.

11.4 Revision notes

Table 30. Revision history

Rev	Date	Change
1.0	2026-10-08	Initial release.